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Dynamic Historical Data-Based Reinforcement Learning for Pursuit–Evasion Games of Nonholonomic Vehicles With Input Saturation

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***Abstract*—This article studies the pursuit–evasion game involving nonholonomic vehicles constrained by input saturation, aiming for the pursuer to intercept an evasive opponent. Unlike the previous game research neglecting the practical kinematic constraints, a coupled nonlinear system is formulated to elucidate the interaction dynamics between the players. After that, the optimal control strategies are derived by solving the Hamilton–Jacobi–Isaacs (HJI) equation linked to a special nonquadratic cost function. The Nash equilibrium analysis and finite-time capturability are conducted. To learn the optimal pursuit–evasion strategy pair, a fixed-time convergent reinforcement learning (RL) algorithm is proposed, which leverages a novel residual design to facilitate weight updates by collecting and evaluating current and historical data based on information quality. Compared with the existing RL methods that suffer from sluggish convergence due to an asymptotic learning rule and the stringent persistent excitation (PE) condition, the proposed RL relaxes the PE to an easily achievable and online verifiable finite excitation (FE) condition, allowing rapid weight convergence within a fixed period. Simulations and comparisons validate the effectiveness and superiority of the proposed method, showing a 61% reduction in convergence time in contrast to the prevailing RL schemes.**

***Index Terms*—Finite excitation (FE), nonholonomic vehicles, pursuit–evasion games, reinforcement learning (RL), saturating actuators.**

I. Introduction

With the advancement of computer technology, communication networks, and sensor hardware, autonomous vehicles are playing increasingly important roles in practical applications, such as area surveillance, search and rescue, and logistics transportation [1], [2], [3], [4]. Different from the above missions, research on adversarial games involving autonomous vehicles is gradually becoming an attractive focus as the level of intelligence increases for systems. In particular, inspired by the hunting activities observed in the animal kingdom, the pursuit–evasion game has garnered significant attention as an intriguing study topic [5], [6], [7]. For instance, in the Russia–Ukraine war, both sides deploy suicide autonomous vehicles carrying munitions to effectively strike strategic targets, resembling a pursuit–evasion game scenario [8]. Typically, in this game, considering both parties possess rational decision-making abilities, the pursuer's objective is to exert maximal effort to track and capture the evader, while the evader aims to evade being tracked as much as possible. Therefore, it can be regarded as a subcategory of differential game theory, specifically a zero-sum (ZS) game, whose total revenue is zero throughout this process [9], [10].

Initially, drawing from the conflict and optimization feature between the two players, Isaacs [11] studied the pursuit–evasion issue based on control theory, which is designed to reveal self-interested behavior and derive the optimal policies of competitive agents interacting in a dynamic changing environment. Later, Lewis et al. [12] extended the Bellman equation as the Hamilton–Jacobi–Isaacs (HJI) equations to construct H_∞ control strategy based on the ZS game. Utilizing this foundation, fruitful investigations on ZS games have been accomplished for linear systems [13], [14], in which the control objectives focus on minimizing the maximum norm as regards control actions and system states. Moreover, for the multiagent system, a graph-theoretic-based distributed optimal control strategy is proposed to minimize the given performance indices in [15]. The sufficient conditions for implementing the exponential capture are revealed for multiagent pursuit–evasion game in [16]. However, the works in [13], [14], [15], and [16] overlook the nonlinear dynamics, rendering them difficult to directly apply to vehicles. The primary challenge in addressing nonlinear ZS games lies in the difficulty of obtaining analytical forms of the Nash equilibrium solutions due to the intractable nonlinearity and partial differential equations.

Reinforcement learning (RL) has been identified as a viable method to approximate the solution of HJI equation for nonlinear systems [17], [18], [19], [20]. The design kernel is dedicated to inferring the near-optimal control strategy pair of both players based on the neural network (NN) approximator. Accordingly, it circumvents the challenges encountered in

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classical optimal control approaches. An integral RL-based optimal controller is developed to plan the capture path of autonomous aerial vehicles in [21]. Through online learning the value function, the Nash equilibrium of the pursuit–evasion game is obtained for agents in [22]. Based on the linearized relative motion equation, the differential game of unmanned surface vehicles is studied in [23], where a single critic and two actors are utilized to update the optimal cost and optimal policies of both players, respectively. Moreover, a critic-only learning algorithm is proposed to learn the game policies for the satellites with input saturation based on simplified linear kinematics in [24].

It is noteworthy that despite the favorable progress made in optimal pursuit–evasion strategies, several challenges remain unresolved. First, the existing literature [21], [22], [23], [24] often simplifies the vehicle model to a linear system, which may hamper the accurate prediction concerning the evader's decision. Second, the current results [21], [22], [23] focus on optimizing linear quadratic cost functions, neglecting the practical actuator saturation, such as the forward velocity and angular rate limitations of vehicles at the guidance level [25], [26]. Consequently, these strategies struggle to precisely capture the actual characteristics of the systems. Third, most existing RL algorithms use gradient descent to update NN weights, but ensuring convergence is challenging. It relies on the persistent excitation (PE) condition [27], which is hard to verify and meet in practical systems due to the need for continuous input excitation in all directions [28]. Additionally, asymptotic convergence rules [19], [20], [21], [22], [23], [24] make it difficult to achieve satisfactory learning performance. To relax the PE requirement, experience replay techniques [29], [30] use stored data for NN updates, but data selection and information matrix quality remain problematic. While a finite-time safe pursuit–evasion problem is addressed in [31] with an adaptive law for finite-time convergence, its reaching time upper bound increases with initial weight error. Note that the fixed-time convergence theory [32] can address this limitation, but relevant research is extremely scarce.

Motivated by the aforementioned considerations, this article studies the pursuit–evasion game problem for vehicles with nonholonomic constraints and input saturation. The objective is to enable the pursuer to predict the evader's optimal strategy while adopting corresponding countermeasures to achieve capture. To this end, we derive the relative motion dynamics and establish a coupled nonlinear system for both players. Then, the constrained pursuit–evasion issue is formulated as an infinite-horizon ZS game problem, and the Nash equilibrium strategies are deduced. Finally, an online RL method is proposed to approximate the solution of the HJI equation. The strict proof of the algorithm's convergence is provided, and the impact of NN approximation errors on the Nash equilibrium points and system states is analyzed. The primary contributions are as follows.

1) A bounded Nash equilibrium policy is derived for the first time by incorporating the players' physical constraints. Compared with the previous pursuit–evasion games [13], [14], [15], [16], [21], [22], [23], [24] that oversimplify player models as linear systems, the proposed scheme accounts for both the nonlinear characteristics arising from nonholonomic constraints and the effects of input saturation, thereby better reflecting realistic interaction behaviors without exaggerating maneuverability.
2) A novel dynamic historical data-based RL algorithm is proposed to approximate the game strategies of both players. This approach relaxes the strict PE condition required in the existing RL methods[19], [20], [21], [22], [23], [24], replacing it with a more easily achievable and online verifiable finite excitation (FE) condition, while ensuring convergence to the Nash equilibrium solution within a fixed time, rather than exhibiting unbounded time in [19], [20], [21], [22], [23], [24], [27], [28], [29], and [30].

The remainder of this article is organized as follows. In Section II, the problem formulation is presented. In Section III, the optimal pursuit–evasion game policies are derived and analyzed. In Section IV, the online RL method is proposed, and its stability is proved. Section V presents the simulations. Section VI concludes this article.

II. Problem Formulation

A. Vehicle Kinematics

Consider a pursuit–evasion game occurring in a compact set $\Omega \subset \mathbb{R}^2$, where a single pursuing vehicle is tasked with capturing an uncooperative evading vehicle on a two-dimensional plane. Each vehicle is modeled as a nonholonomic system expressed in the global coordinate frame as follows:

$$\dot{x}_i = v_i \cos\psi_i,\ \ \dot{y}_i = v_i \sin\psi_i,\ \ \dot{\psi}_i = \omega_i \tag{1}$$

where $i \in \{p, e\}$ with p being the pursuer and e representing the evader. $\xi_i = [x_i, y_i]^T \in \mathbb{R}^2$ is the position, and $\psi_i \in \mathbb{R}$ denotes the heading angle. $v_i \in \mathcal{U}$ and $\omega_i \in \mathcal{W}$ denote the bounded speed and angular rate, which reflect the practical physical constraints in the sense that $\mathcal{U} := \{v_i \in \mathbb{R} : |v_i| \le \bar{v}_i\}$ with $\bar{v}_i \in \mathbb{R}^+$ and $\mathcal{W} := \{\omega_i \in \mathbb{R} : |\omega_i| \le \bar{\omega}_i\}$ with $\bar{\omega}_i \in \mathbb{R}^+$.

Remark 1: To emphasize the design of the optimal maneuvering policy, we model the system dynamics of both vehicles using the kinematic Dubins model and consider low-level autopilot perfectly tracking of kinematic commands. This model is a widely accepted and validated choice for representing vehicle motion [1], [25], [26], as it captures critical constraints, such as nonholonomic properties and bounded control inputs. The combination of these nonlinear dynamics and the strategic interactions between players leads to a nonlinear and coupled HJI equation [12]. This study focuses on deriving and learning the bounded Nash equilibrium strategies from the HJI equation.

B. Pursuit–Evasion Game Formulation

Considering two perfectly rational decision-making vehicles, defined as the players, which can detect the state of the game and always choose to follow their expected optimal policies, where the pursuer aims to minimize the relative distance to the evader, while the evader seeks to maximize it. Thus, this problem can be formulated as a two-player ZS

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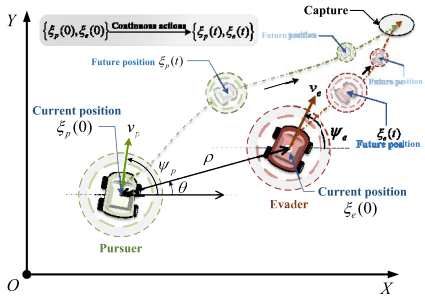


Fig. 1. Illustration of the pursuit–evasion game for nonholonomic vehicles.

game. Note that it is crucial to determine the relative motion during the pursuit. Therefore, instead of deriving the control laws directly in the global coordinates, we handle the relative motion dynamics of the players in the polar coordinates, as shown in Fig. 1. Define the relative range and azimuth angle as

$$\begin{cases} \rho = \|\xi_p - \xi_e\| = \sqrt{(x_p - x_e)^2 + (y_p - y_e)^2} \\ \theta = \text{atan2}\big((y_e - y_p), (x_e - x_p)\big) \end{cases} \tag{2}$$

where atan2 is an arctangent function. Differentiating (2) yields

$$\begin{cases} V_\rho = \dot{\rho} = v_e \cos(\theta - \psi_e) - v_p \cos(\theta - \psi_p) \\ \varrho = \dot{\theta} = [v_e \sin(\psi_e - \theta) - v_p \sin(\psi_p - \theta)]/\rho. \end{cases} \tag{3}$$

Later, we consider that both players maintain the maximum permissible speed during the game, which means that the control input only changes the direction of the speed without altering its magnitude. Taking the time derivative of ϱ renders

$$\dot{\varrho} = -\frac{2\dot{\rho}\varrho}{\rho} - \frac{\omega_p v_p \cos(\psi_p - \theta)}{\rho} + \frac{\omega_e v_e \cos(\psi_e - \theta)}{\rho}. \tag{4}$$

Formulating the augmented state vector as

$$\zeta = [\zeta_1, \quad \zeta_2]^T = [\theta \quad \varrho]^T \in \mathbb{R}^2. \tag{5}$$

renders the following dynamics:

$$\dot{\zeta} = F(\zeta) + G(\zeta)u_p + K(\zeta)u_e, \ \ \zeta(0) = \zeta_0, \ \ t \geq 0 \tag{6}$$

where $u_p \triangleq \omega_p$, and $u_e \triangleq \omega_e$; $F(\zeta) \triangleq [\zeta_2, -(2\dot{\rho}\zeta_2/\rho)]^T$, $G(\zeta) \triangleq [0, -(v_p \cos(\psi_p - \theta)/\rho)]^T$, and $K(\zeta) \triangleq [0, (v_e \cos(\psi_e - \theta)/\rho)]^T$ are Lipschitz continuous and limited by constants G_M and K_M such that $\|G(\zeta)\| \leq G_M$ and $\|K(\zeta)\| \leq K_M$. It can be seen that the pursuit–evasion game essentially revolves around the rate of change of the azimuth angle. The pursuer's goal can be reformulated as stabilizing state ζ to the origin in response to the evader's optimal maneuvers.

Problem 1: Considering the game of two vehicles, the design goal aims to seek the pursuer's optimal action sequence to catch the evader with a perfectly rational decision. Then, one can define the following cost function to evaluate the performance:

$$J(\zeta(0), u_p, u_e) = \int_0^\infty L(\zeta(t), u_p(t), u_e(t))dt \tag{7}$$

where $L(\zeta, u_p, u_e) \triangleq L_\zeta(\zeta) + L_p(u_p) - L_e(u_e)$; $L_\zeta(\zeta) \triangleq \zeta^T Q \zeta$ represents a quadratic function such that one pursuer can approach the evader, with $Q^T = Q > 0$. $L_p(u_p)$ and $L_e(u_e)$ are penalty functions to be designed to characterize bounded inputs. Given that ρ cannot be zero in practical applications, we suppose the existence of a minimum relative range ρ_0, and the mission will end when $\rho < \rho_0$. Then, the pursuer intends to minimize the cost function (7) while the evader aims to maximize it. Hence, it can be formulated as the constrained ZS game problem

$$\begin{aligned} &V^*(\zeta(t)) = \min_{u_p} \max_{u_e} \int_t^\infty L(\zeta(\tau), u_p(\tau), u_e(\tau))d\tau \quad \forall t \geq 0 \\ &\text{s.t.} \ \ |v_i| \leq \bar{v}_i, \ \ |u_i| \leq \bar{\omega}_i, \ \ i \in \{p, e\} \\ &\qquad (6), \ \ \zeta(0) = \zeta_0 \end{aligned} \tag{8}$$

where $V^*(\zeta(t))$ represents the optimal value function of the system (6). Then, the design objective is to find a stabilizing saddle point solution, such that the equilibrium point of the nonlinear system (6) is stable while $(u_p^*(\cdot), u_e^*(\cdot))$ represents the saddle point of the given performance index (7) for any ζ_0.

Before proceeding, we make the following assumption.

Assumption 1: At $t = 0$, the pursuing vehicle can observe the state of the target and does not face the issue of initially locating it. The maximum velocity of the pursuer satisfies $\bar{v}_p > \bar{v}_e$, and the interaction occurs in an open, obstacle-free environment without external spatial constraints.

Remark 2: Assumption 1 provides a conservative condition adopted in classical pursuit–evasion games [23], [24] to ensure that the single pursuer can intercept the evader in free space. It indicates that the pursuer's maximum speed must exceed that of the evader; otherwise, if the evader maintains its maximum velocity, the pursuer cannot close the relative distance to the target. Although certain scenarios, such as those involving confined domains or external obstacles, may enable the slower pursuer to capture a faster evader, the analysis in such cases becomes significantly more complex due to the influence of environmental constraints. Therefore, we assume that the game environment is free from external constraints.

Remark 3: Compared to the existing studies [13], [14], [15], [16], [21], [22], [23], [24] that use an oversimplified integrator model without involving the physical dynamics and input constraints, the relative motion dynamics model (6) for the two kinematic vehicles is derived as a nonlinear system. This approach is more practical and produces outcomes that are more consistent with actual circumstances. Furthermore, under the assumption that players maintain maximum speed, the upper bound on angular rate can be derived based on the turning radius, providing a reasonable reflection of vehicle capabilities.

III. Optimal Pursuit–Evasion Game Strategies Design

In this section, we first utilize the Bellman optimality principle to derive the optimal strategy pair for the ZS game. Second, the criteria for Nash equilibrium of the game are established and the conditions of successful capture are

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derived, while the complex interactions between the players are revealed.

A. Differential Game Between the Pursuer and Evader

To ensure that the inputs of both players remain within specified bounds, we define nonquadratic penalty functions to realize the input constraints, i.e.,

$$L_p(u_p) = 2\int_0^{u_p} (\alpha^{-1}(v/\lambda_p))R_p dv \tag{9}$$

$$L_e(u_e) = 2\int_0^{u_e} (\alpha^{-1}(\varpi/\lambda_e))R_e d\varpi \tag{10}$$

where $R_i \in \mathbb{R}^+$ is a positive value for $i = p, e$, and it plays a key role by confining the change rate of optimal policy to constrain the accelerations of players; $\alpha(\cdot)$ means a limited one-to-one monotonic odd function and its differential is bounded by a known constant. One can select $\tanh(\cdot)$ as an alternative, which can confine the control input to maintain the predefined constraints, i.e., $|u_i| \le \lambda_i, \forall i = p, e$.

After designing the utility function $L(\zeta, u_p, u_e)$, we further study the optimal solution. The saddle-point conditions satisfy

$$J(\zeta, u_p^*, u_e^*) = \max_{u_e(\cdot)} J(\zeta, u_p^*, u_e) = \min_{u_p(\cdot)} J(\zeta, u_p, u_e^*) \quad \forall \zeta. \tag{11}$$

The term $J(\zeta, u_p^*, u_e^*)$ is claimed as the optimal value function

$$V^*(\zeta) \triangleq J(\zeta, u_p^*, u_e^*) \quad \forall \zeta. \tag{12}$$

Then, by computing the derivative of the functional $V(\zeta) \triangleq \int_t^\infty \{L_\zeta(\zeta) + L_p(u_p(\zeta)) - L_e(u_e(\zeta))\}d\tau$ based on the system (6) and Leibniz's formula, we can construct the corresponding Hamiltonian function as

$$H(\zeta, \nabla V(\zeta), u_p, u_e) = L_\zeta(\zeta) + L_p(u_p) - L_e(u_e) + \nabla^T V(\zeta) \times (F(\zeta) + G(\zeta)u_p + K(\zeta)u_e) = 0 \tag{13}$$

where $\nabla V(\zeta) = \partial V/\partial\zeta \in \mathbb{R}^2$ is the gradient of $V(\zeta)$.

Next, to seek the optimal strategies u_p^* and u_e^* of two players, one can set stationarity conditions $\partial H/\partial u_p = 0$ and $\partial H/\partial u_e = 0$. Thus, for the pursuer, employing Pontryagin's minimum criterion as regards (13) renders

$$\frac{\partial H}{\partial u_p} = \frac{\partial L_p}{\partial u_p} + \frac{\partial}{\partial u_p}\{\nabla^T V(\zeta)(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e)\}. \tag{14}$$

Then, (14) can be solved as

$$2\left(\tanh^{-1}\left(\frac{u_p^*}{\lambda_p}\right)\right)R_p = -\nabla^T V^* G(\zeta). \tag{15}$$

The optimal policy of pursuing players can be derived as

$$u_p^* \triangleq \arg\min_{u_p} H(\zeta, \nabla V(\zeta), u_p, u_e) = -\lambda_p \tanh\left(\frac{1}{2}R_p^{-1}G^T(\zeta)\nabla V^*(\zeta)\right). \tag{16}$$

Similarly, the bounded optimal policy of evading players can be obtained as

$$u_e^* \triangleq \arg\max_{u_e} H(\zeta, \nabla V(\zeta), u_p, u_e) = \lambda_e \tanh\left(\frac{1}{2}R_e^{-1}K^T(\zeta)\nabla V^*(\zeta)\right). \tag{17}$$

From the fact of $\int \tanh^{-1}\tau d\tau = \tau\tanh^{-1}\tau + (1/2)\ln|1-\tau^2|$, as well as combining (9) and (16), one has

$$L_p(u_p^*) = 2R_p\left[u_p^*\tanh^{-1}\left(\frac{u_p^*}{\lambda_p}\right) + \frac{\lambda_p}{2}\ln\left(1-\left(\frac{u_p^*}{\lambda_p}\right)^2\right)\right] = \lambda_p\nabla^T V^*(\zeta)G\tanh(\Phi_p^*) + \lambda_p R_p\ln\left(1-\tanh^2(\Phi_p^*)\right) \tag{18}$$

where $\Phi_p^* = (1/2)R_p^{-1}G^T(\zeta)\nabla V^*(\zeta)$. Likewise, one has

$$L_e(u_e^*) = \lambda_e\nabla^T V^*(\zeta)K\tanh(\Phi_e^*) + \lambda_e R_e\ln\left(1-\tanh^2(\Phi_e^*)\right) \tag{19}$$

where $\Phi_e^* = (1/2)R_e^{-1}K^T(\zeta)\nabla V^*(\zeta)$.

Plugging (16) and (17) into (13), one can get the HJI equation

$$H(\zeta, \nabla V^*, u_p^*, u_e^*) = \max_{u_e}\min_{u_p} H(\zeta, \nabla V^*, u_p, u_e) = \zeta^T Q\zeta + L_p(u_p^*) - L_e(u_e^*) + \nabla^T V^*(\zeta) \times (F(\zeta) + G(\zeta)u_p^* + K(\zeta)u_e^*) = 0. \tag{20}$$

Thus, we can observe that the pursuer's strategy is to minimize the Hamiltonian (13) whereas the evader's is to maximize it, and policy pair (u_p^*, u_e^*) is a unique saddle point.

B. Nash Equilibrium

This section aims to deduce the optimal value of the two-player pursuit–evasion game at Nash equilibrium.

Theorem 1: Consider the relative motion dynamics (6) of the players (1) with value function $V(\zeta)$. Define $V(\zeta)$ as the solution of the Hamiltonian (13) and let $V^*(\zeta)$ denote a positive definite solution as regards HJI equation (20). Then, (u_p^*, u_e^*) provided by (16) and (17) is the Nash equilibrium of pursuit–evasion game while $V^*(\zeta)$ denotes the related optimal value.

Proof: First, adding and subtracting $L_p(u_p^*), L_e(u_e^*)$, $\nabla^T V(\zeta)G(\zeta)u_p^*$, and $\nabla^T V(\zeta)K(\zeta)u_e^*$ to (13) leads to $H(\zeta, \nabla V(\zeta), u_p, u_e) = L_\zeta(\zeta) + L_p(u_p^*) - L_e(u_e^*) + \nabla^T V(\zeta)(F(\zeta) + G(\zeta)u_p^* + K(\zeta)u_e^*) + \nabla^T V(\zeta)(G(\zeta)(u_p - u_p^*) + K(\zeta)(u_e - u_e^*)) + L_p(u_p) - L_p(u_p^*) - L_e(u_e) + L_e(u_e^*) = H(\zeta, \nabla V(\zeta), u_p^*, u_e^*) + \nabla^T V(\zeta)(G(\zeta)(u_p - u_p^*) + K(\zeta)(u_e - u_e^*)) + L_p(u_p) - L_p(u_p^*) - L_e(u_e) + L_e(u_e^*)$.

Then, let $V(\zeta(t_0))$ be the initial value of the game. Suppose that the capture behavior of players happens in $t \in [t_0, \infty]$, manifesting $\lim_{t\to\infty} V(\zeta(t)) = 0$. One has

$$\begin{aligned} V(\zeta(t_0)) &= \int_{t_0}^\infty \{L_\zeta(\zeta) + L_p(u_p(\zeta)) - L_e(u_e(\zeta))\}\, d\tau \\ &\quad + \int_{t_0}^\infty \dot{V}(\zeta)\, d\tau + V(\zeta(t_0)) \\ &= \int_{t_0}^\infty \{L_\zeta(\zeta) + L_p(u_p) - L_e(u_e)\}\, d\tau + V_0(\zeta(t_0)) \\ &\quad + \int_{t_0}^\infty \nabla V(\zeta)\big(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e\big)\, d\tau \\ &= \int_{t_0}^\infty H(\zeta, \nabla V(\zeta), u_p, u_e)\, d\tau + V(\zeta(t_0)). \end{aligned} \tag{21}$$

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Next, we assume that $V^*(\zeta)$ meets the HJI equation (20), one gets $H(\zeta, \nabla V^*, u_p^*, u_e^*) = 0$. Thus, it yields

$$H(\zeta, \nabla V^*, u_p, u_e) = \nabla^T V^*(\zeta)(G(\zeta)(u_p - u_p^*) + K(\zeta) \times (u_e - u_e^*)) + L_p(u_p) - L_p(u_p^*) - L_e(u_e) + L_e(u_e^*). \quad (22)$$

Later, substituting (22) into (21), we can find

$$V(\zeta(t)) = \int_t^\infty \{\nabla^T V^*(\zeta)(G(\zeta)(u_p - u_p^*) + K(\zeta)(u_e - u_e^*)) + L_p(u_p) - L_p(u_p^*) - L_e(u_e) + L_e(u_e^*)\} d\tau + V^*(\zeta(t_0)). \quad (23)$$

To demonstrate that the policy pair (u_p^*, u_e^*) denotes the Nash equilibrium, it is critical to reveal when the pursuer takes action provided by (16), the evader's optimal behavior that maximizes the value function $V(\zeta)$ is (17), and vice versa. Thus, it yields

$$V(\zeta, u_p^*, u_e) \le V(\zeta, u_p^*, u_e^*) \le V(\zeta, u_p, u_e^*). \quad (24)$$

Based on (23), we can derive that $V(\zeta, u_p^*, u_e^*) = V^*(\zeta(t_0))$. Then, letting $\chi(V(\zeta, u_p, u_e)) \triangleq \int_t^\infty \{\nabla^T V^*(\zeta)(G(\zeta)(u_p - u_p^*) + K(\zeta)(u_e - u_e^*)) + L_p(u_p) - L_p(u_p^*) - L_e(u_e) + L_e(u_e^*)\} d\tau$. Then it requires revealing $\chi(V(\zeta, u_p^*, u_e)) \le 0$ and $\chi(V(\zeta, u_p, u_e^*)) \ge 0$ such that (24) holds. Using (9), (10), (16), (17), and (23), we get

$$\begin{aligned}\chi(V(\zeta, u_p^*, u_e)) &= \int_t^\infty \left\{\nabla^T V^*(\zeta)K(\zeta)(u_e - u_e^*) - L_e(u_e) + L_e(u_e^*)\right\} d\tau \\ &= \int_t^\infty \left\{2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e - u_e^*) + 2\int_{u_e}^{u_e^*}\left(\tanh^{-1}\left(\frac{\varpi}{\lambda_e}\right)\right)R_e d\varpi\right\} d\tau. \quad (25)\end{aligned}$$

Note that $\tanh^{-1}(\cdot)$ satisfies the monotonically increasing property in $[-\lambda_e, \lambda_e]$, and we first suppose that $u_e \le u_e^*$. Using the integral mean value theorem renders

$$\chi(V(\zeta, u_p^*, u_e)) \le \int_t^\infty \left\{2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e - u_e^*) - 2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e - u_e^*)\right\} d\tau = 0. \quad (26)$$

Then we suppose that $u_e \ge u_e^*$, and one has

$$\begin{aligned}\chi(V(\zeta, u_p^*, u_e)) &= \int_t^\infty \left\{-2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e^* - u_e) - 2\int_{u_e^*}^{u_e}\left(\tanh^{-1}\left(\frac{\varpi}{\lambda_e}\right)\right)R_e d\varpi\right\} d\tau \\ &\le \int_t^\infty \left\{-2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e^* - u_e) - 2\left(\tanh^{-1}\left(\frac{u_e^*}{\lambda_e}\right)\right)R_e(u_e - u_e^*)\right\} d\tau = 0. \quad (27)\end{aligned}$$

which proves that $\chi(V(\zeta, u_p^*, u_e)) \le 0$. Similarly, one can conclude that $\chi(V(\zeta, u_p, u_e^*)) \ge 0$. Finally, one gets $V(\zeta, u_p^*, u_e) = \chi(V(\zeta, u_p^*, u_e)) + V^*(\zeta(t_0)) = \chi(V(\zeta, u_p^*, u_e)) + V(\zeta, u_p^*, u_e^*) \le V(\zeta, u_p^*, u_e^*)$, and $V(\zeta, u_p, u_e^*) = \chi(V(\zeta, u_p, u_e^*)) + V^*(\zeta(t_0)) = \chi(V(\zeta, u_p, u_e^*)) + V(\zeta, u_p^*, u_e^*) \ge V(\zeta, u_p^*, u_e^*)$, which manifests that the inequality (24) maintains true. Thus, (u_p^*, u_e^*) is the Nash equilibrium while $V^*(\zeta(t_0))$ denotes the optimal value of the game. ■

C. Capture Analysis

After proving the Nash equilibrium in the pursuit–evasion game, we will elaborate sufficient conditions for the pursuer to capture the evader during the pursuit process, and the stability of the whole system is certified by using the Lyapunov tool.

Theorem 2: Consider the dynamics (6) and value function (8) for players with input constraints. The equilibrium of the system (6) is asymptotically stable by setting suitable parameters.

Proof: Given a Lyapunov function as $V_1(\zeta) = V(\zeta, u_p^*, u_e^*)$, its differential is $\dot{V}_1 = \nabla^T V_1 \dot{\zeta}$. Using (13), (18), and (19), one has

$$\begin{aligned}\dot{V}_1(\zeta) =& -\zeta^T Q\zeta - 2\int_0^{u_p^*}(\tanh^{-1}(\nu/\lambda_p))R_p d\nu + 2\int_0^{u_e^*}(\tanh^{-1}(\varpi/\lambda_e))R_e d\varpi \\ =& -\zeta^T Q\zeta - \left[\lambda_p \nabla^T V^*(\zeta) G \tanh(\Phi_p^*) + \lambda_p R_p \ln(1 - \tanh^2(\Phi_p^*))\right] + \left[\lambda_e \nabla^T V^*(\zeta) K \tanh(\Phi_e^*) + \lambda_e R_e \times \ln\left(1 - \tanh^2(\Phi_e^*)\right)\right] \\ =& -\zeta^T Q\zeta - \left[\lambda_p \nabla^T V^*(\zeta) G \tanh(\Phi_p^*) - \lambda_e \nabla^T V^*(\zeta) \times K \tanh(\Phi_e^*)\right] - \left[\lambda_p R_p \ln\left(1 - \tanh^2(\Phi_p^*)\right) - \lambda_e R_e \ln\left(1 - \tanh^2(\Phi_e^*)\right)\right]. \quad (28)\end{aligned}$$

Note that at the moment when the pursuer is about to capture the evader, the pursuer's heading angle ψ_p should align with the azimuth angle θ. If the evader makes a sudden maneuver, i.e., changing its angular velocity u_e^*, the pursuer must be capable of executing a larger maneuver to adapt to the evader's changes. Therefore, asymptotic capture occurs under the optimal strategy u_p^* satisfying $|u_p^*| > |u_e^*|$, i.e., $\lambda_p > \lambda_e$. Moreover, for $\dot{V}_1(\zeta)$ to be negative definite, one requires $\lambda_p \nabla^T V^*(\zeta) G \tanh(\Phi_p^*) > \lambda_e \nabla^T V^*(\zeta) K \tanh(\Phi_e^*)$ and $\lambda_p R_p \ln(1 - \tanh^2(\Phi_p^*)) > \lambda_e R_e \ln(1 - \tanh^2(\Phi_e^*))$. Since $\tanh(\cdot)$ is limited, one has $\tanh^2(\Phi_p^*) \le 1$ and $\tanh^2(\Phi_e^*) \le 1$, yielding that $\ln(1 - \tanh^2(\Phi_p^*))$ and $\ln(1 - \tanh^2(\Phi_e^*))$ are negative. Moreover, the sign of $\tanh(\mathfrak{D})$ is consistent with the sign of the variable $\mathfrak{D}$. Then, one can set $R_e > R_p$ such that $\dot{V}_1$ can be negative definite. Thus, the equilibrium of the system dynamics (6) is asymptotically stable. ■

Now, consider the areas occupied by the pursuer and evader with a collision radius denoted as ρ_0. Capture is achieved when the relative range between players is less than ρ_0. Then, the following main theorem demonstrates that the pursuer successfully captures the evader within a finite settling time.

Theorem 3: Provided the Lyapunov function $V_1(\zeta)$, the system (6) can converge to the origin. Meanwhile, one can

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seek an upper bound concerning the capture time, meaning that the pursuit–evasion game terminates within a finite time.

Proof: First, we analyze the properties of $V_1(\zeta)$. Based on Theorem 2, one obtains that $L_p(u_p^*(\zeta)) \geq L_e(u_e^*(\zeta))$. Then, there exist class $\mathcal{K}_\infty$ functions $\underline{c}, \bar{c} \in \mathbb{R}^+$, such that $\underline{c}(\|\zeta\|) \leq V_1(\zeta) \leq \bar{c}(\|\zeta\|)$. Let the convex function $L_c(\zeta)$ satisfy $L_c(\zeta) \leq L_p(u_p^*(\zeta)) - L_e(u_e^*(\zeta))$. Based on (28), one has

$$\dot{V}_1(\zeta) \leq -\zeta^T Q \zeta - L_c(\zeta) \tag{29}$$

which denotes that $\dot{V}_1(\zeta) \leq -c_1 V_1(\zeta)$ with c_1 being a sufficiently small constant. Substituting (29) into $V^*(\zeta)$, it can derive

$$\|\zeta(t)\| \leq b_1 \|\zeta(t_0)\| e^{-b_2(t-t_0)} \quad \forall t > t_0 \tag{30}$$

where b_1 and b_2 are positive coefficients. Then, instead of directly using the physical capture condition $\rho = \rho_0$, we calculate the time t_f required for the state norm to converge to a small threshold $\epsilon_\zeta > 0$. Achieving this stable guidance condition, i.e., $\|\zeta(t_f)\| \leq \epsilon_\zeta$, is a necessary precursor to the final capture. Making a transition for (30) yields $(\|\zeta(t_f)\|/b_1\|\zeta(t_0)\|) \leq e^{-b_2(t_f-t_0)}$. Taking the logarithm of both sides results in $t_f - t_0 \leq (\ln(b_1\|\zeta(t_0)\|/\|\zeta(t_f)\|)/b_2)$. Next, by substituting the terminal condition, we obtain an upper bound for the time to achieve this guidance state

$$t_f \leq t_0 + \ln\big(b_1\|\zeta(t_0)\|/\epsilon_\zeta\big)/b_2 \tag{31}$$

which completes the proof. ■

Remark 4: Different from the existing pursuit–evasion games [21], [22], [23], [24], [31] that neglect limited control input, this article demonstrates the Nash equilibrium for pursuit–evasion games of vehicles subject to actuator saturation. It reveals that the optimal strategy pair (u_p^*, u_e^*) forms a unique saddle point. Moreover, Theorems 2 and 3 further certify finite-time capture by considering the practical existence of a capture radius, leading to the significant conclusion that the initial conditions between the players do not affect the success of the capture.

Based on Theorems 1–3, achieving the capture for the pursuer necessitates solving the ZS game problem (8), which requires finding the solution to the HJI equation (20). However, (20) is a nonlinear partial differential equation, making it challenging to solve analytically. To address this challenge, the online RL will be implemented to obtain an approximated solution.

IV. Online Approximation Solution Using Reinforcement Learning Without PE

In this section, a simplified RL algorithm is presented to online approximate the solution to HJI equation in a fixed time without requiring the PE condition. The detailed implementation diagram is shown in Fig. 2.

A. Design of Critic NN

Based on the Weierstrass approximation theorem, $V^*(\zeta)$ can be locally approximated via an NN approximator for $\zeta \in \mathcal{X}$ with $\mathcal{X}$ being a compact set [21], which can be formalized by

$$V^*(\zeta) = W^T \Psi(\zeta) + \epsilon(\zeta) \tag{32}$$

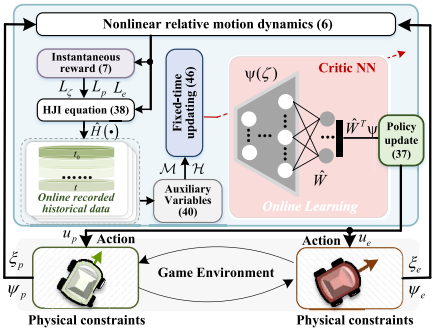


Fig. 2. FE-driven online RL algorithm for the pursuit–evasion game of vehicles.

and the gradient of $V^*(\zeta)$ is

$$\nabla V^*(\zeta) = \nabla^T \Psi(\zeta) W + \nabla \epsilon(\zeta) \tag{33}$$

where $W = [W_1, W_2, \ldots, W_N]^T \in \mathbb{R}^N$ is the ideal weight vector, constrained by $\|W\| \leq \bar{W}$; $\Psi(\zeta) = [\Psi_1(\zeta), \Psi_2(\zeta), \ldots, \Psi_N(\zeta)]^T \in \mathbb{R}^N$ represents the nonlinear activation function; N stands for the number of neurons; $\epsilon(\zeta) \in \mathbb{R}$ is the bounded error. Gradients $\nabla\Psi = \partial\Psi/\partial\zeta \in \mathbb{R}^{N\times 2}$ and $\nabla\epsilon = \partial\epsilon/\partial\zeta \in \mathbb{R}^2$ satisfy $\|\nabla\Psi\| \leq \bar{\Psi}$ and $|\nabla\epsilon| \leq \bar{\epsilon}$ [33].

Utilizing (33) and given a fixed control action pair (u_p, u_e), the Hamiltonian function becomes

$$H(\zeta, W, u_p, u_e) = L_\zeta(\zeta) + L_p(u_p) - L_e(u_e) + W^T \nabla\Psi(\zeta) \times (F(\zeta) + G(\zeta)u_p + K(\zeta)u_e) = \varepsilon_H \tag{34}$$

where $\varepsilon_H = -\nabla^T \epsilon(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e)$ is the residual error, which is bounded and can be sufficiently small via employing numerous neurons [29].

Inserting (33) into (16) and (17), the pursuer's optimal policy $u_p^*(\zeta)$ becomes

$$u_p^*(\zeta) = -\lambda_p \tanh\left(\frac{1}{2} R_p^{-1} G^T(\zeta)(\nabla^T \Psi(\zeta) W + \nabla\epsilon(\zeta))\right) \tag{35}$$

while the evader's optimal policy is

$$u_e^*(\zeta) = \lambda_e \tanh\left(\frac{1}{2} R_e^{-1} K^T(\zeta)(\nabla^T \Psi(\zeta) W + \nabla\epsilon(\zeta))\right). \tag{36}$$

Since W cannot be exactly determined, we utilize the estimated NN weights $\hat{W} \in \mathbb{R}^N$ to approximate the value of $V^*(\zeta)$, i.e., $V(\zeta, \hat{W}) \triangleq \hat{W}^T \Psi(\zeta)$, yielding that

$$\begin{cases} u_p(\zeta, \hat{W}) \triangleq -\lambda_p \tanh\left(\frac{1}{2} R_p^{-1} G^T(\zeta) \nabla^T \Psi(\zeta) \hat{W}\right) \\ u_e(\zeta, \hat{W}) \triangleq \lambda_e \tanh\left(\frac{1}{2} R_e^{-1} K^T(\zeta) \nabla^T \Psi(\zeta) \hat{W}\right). \end{cases} \tag{37}$$

B. Design of Fixed-Time Updating Law Without PE

In this section, we introduce a novel fixed-time updating law for enhancing the convergence of NN weights while eliminating PE. The design objective is to seek an online learning rule for the weight estimates $\hat{W}$, and thus offer a good approximation

$$\hat{H}(\zeta, \hat{W}^T \nabla\Psi, u_p, u_e) \triangleq \hat{W}^T \nabla\Psi(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e) + L_\zeta(\zeta) + L_p(u_p) - L_e(u_e) \tag{38}$$

for the Hamiltonian.

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Letting $\sigma_1 = \nabla\Psi(\zeta)(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e)$. Then, define the estimation error of weights as $\tilde{W} = W - \hat{W}$. Subtracting (34) from (38) yields

$$\begin{aligned}&\hat{H}(\zeta, \hat{W}^T\nabla\Psi, u_p, u_e) - H(\zeta, W^T\nabla\Psi, u_p, u_e)\\ &= -\tilde{W}^T\nabla\Psi(F(\zeta) + G(\zeta)u_p + K(\zeta)u_e) + \varepsilon_H \triangleq \delta_e \quad (39)\end{aligned}$$

where δ_e denotes the Hamiltonian residual.

Definition 1 (PE [34]): A vector or matrix Θ is claimed as meeting PE if there exist $T > 0$ and $\kappa > 0$ such that $\int_t^{t+T} \Theta(\tau)\Theta^T(\tau)\,d\tau \geq \kappa I, \forall t \geq 0$.

Definition 2 (FE): A bounded vector or matrix Θ is claimed as satisfying FE over an interval $[T_s, T_s + T]$, if there exist $T > 0$ and $T_s \geq 0$ and $\kappa > 0$ such that $\int_{T_s}^{T_s+T} \Theta(\tau)\Theta^T(\tau)\,d\tau \geq \kappa I$.

Assumption 2 (FE of Regressor): There exist $T_s \geq 0$ and $T_e > T_s$ yielding that the regressor $\sigma_1(t)$ satisfies FE over $[T_s, T_e]$.

Note that Assumption 2 is less restrictive compared to PE. Next, we introduce a novel residual design method for constructing the fixed-time updating rule of NNs based on the FE.

Based on (39), we define the auxiliary matrix and variable as

$$\begin{cases}\dot{\mathcal{H}}(t) &= -\mathcal{K}(t)\ \mathcal{H}(t) + \sigma_1(t)\sigma_1^T(t), \quad \mathcal{H}(0) = 0^{N\times N}\\ \dot{\mathcal{M}}(t) &= -\mathcal{K}(t)\mathcal{M}(t) + \sigma_1(t)L(t), \quad \mathcal{M}(0) = 0^N\\ \mathcal{K}(t) &= k_0\ell(t)\end{cases} \quad (40)$$

where $k_0 > 0$. $\mathcal{K}(t)$ denotes a non-negative time-varying forgetting factor that ensures the boundedness of the matrix $\mathcal{H}(t)$ and vector $\mathcal{M}(t)$. $\ell(t)$ is a smooth scheduling function defined as

$$\ell(t) = \begin{cases}0, & \delta \leq \underline{\delta}\\ 1 - \cos\left(\frac{\pi}{2}\sin\left(\frac{\pi}{2}\frac{\delta^2 - \underline{\delta}^2}{\overline{\delta}^2 - \underline{\delta}^2}\right)\right), & \text{otherwise}\\ 1, & \delta \geq \overline{\delta}\end{cases} \quad (41)$$

in which $\delta \triangleq \lambda_{\min}(\mathcal{H}(t))$ is the minimum eigenvalue of the $\mathcal{H}(t)$; $\underline{\delta}$ and $\overline{\delta}$ are the positive constants that constrain the excitation strength δ.

Then, the solution of (40) is written as

$$\begin{cases}\mathcal{H}(t) &= \int_0^t \exp\left(-\int_\tau^t \mathcal{K}(r)\,dr\right)\sigma_1(\tau)\sigma_1^T(\tau)\,d\tau\\ \mathcal{M}(t) &= \int_0^t \exp\left(-\int_\tau^t \mathcal{K}(r)\,dr\right)\sigma_1(\tau)L(\tau)\,d\tau.\end{cases} \quad (42)$$

Multiplying each side of the equation of (34) by σ_1, one has

$$\sigma_1 L = -\sigma_1\sigma_1^T W + \sigma_1\varepsilon_H. \quad (43)$$

Thus, we can deduce

$$\mathcal{M}(t) = -\mathcal{H}(t)W + \widetilde{\varepsilon}_H \quad (44)$$

where $\widetilde{\varepsilon}_H \triangleq \int_0^t \exp(-\int_\tau^t \mathcal{K}(r)\,dr)\sigma_1(\tau)\varepsilon_H(\tau)\,d\tau \in \mathbb{R}^N$ is the bounded reconstruction error meeting $\|\widetilde{\varepsilon}_H\| \leq \bar{\varepsilon}_H$ with $\bar{\varepsilon}_H$ being a non-negative constant based on the boundedness of $\nabla\Psi$ and $\nabla\epsilon$, and $\widetilde{\varepsilon}_H \to 0$ as $N \to \infty$.

Subsequently, to determine the weight estimation errors $\tilde{W}$, a vector $\mathcal{G}(t) \in \mathbb{R}^N$ is constructed as

$$\mathcal{G}(t) = \mathcal{M}(t) + \mathcal{H}(t)\hat{W} = -\mathcal{H}(t)\tilde{W} + \widetilde{\varepsilon}_H. \quad (45)$$

Finally, a fixed-time updating law with FE is presented as

$$\dot{\hat{W}} = -\Gamma_W\left(\frac{k_{W1}\mathcal{H}^T(t)\mathcal{G}(t)}{\sqrt{\|\mathcal{G}(t)\|^2 + \sigma_W^2}} + k_{W2}\mathcal{H}^T(t)\mathcal{G}(t)\mathcal{G}^T(t)\mathcal{G}(t)\right) \quad (46)$$

where $\Gamma_W \in \mathbb{R}^{N\times N}$ is a positive diagonal gain matrix; $k_{W1} > 0$, $k_{W2} > 0$, and $\sigma_W > 0$ stand for the tuning coefficients.

C. Stability Analysis

This section focuses on analyzing the convergence of weight errors and the impact of NN approximation errors on the Nash equilibrium points and system states using the Lyapunov tool.

Lemma 1 [35]: Consider the real symmetric matrices $A, B \in \mathbb{R}^{n\times n}$. The following inequalities hold: $\lambda_{\min}(A) + \lambda_{\min}(B) \leq \lambda_{\min}(A + B) \leq \lambda_{\max}(A) \leq +\lambda_{\max}(B)$.

Lemma 2 [36]: For any constant $\hbar > 0$ and variable $x \in \mathbb{R}$, the fact of $0 \leq |x| < \hbar + (x^2/\sqrt{x^2 + \hbar^2})$ remains true.

Lemma 3 [32]: If a Lyapunov candidate $V_L(x)$ fulfills $\dot{V}_L(x) \leq -\mu_1 V_L(x)^\alpha - \mu_2 V_L(x)^\beta$, with $\mu_1, \mu_2, \alpha, \beta$ being positive parameters and $\alpha \in (0, 1)$, $\beta \in (1, \infty)$, then the related system is fixed-time stable, and the decaying time T_s is provided by $T_s \leq (1/\mu_1(1 - \alpha)) + (1/\mu_2(\beta - 1))$.

Theorem 4: If the FE for regressor σ_1 can be fulfilled, then one has: 1) $\mathcal{H}(t) \geq 0$ for $\forall t \geq 0$; 2) $\mathcal{H}(t) > 0$ for $\forall t \geq T_e$; and 3) $\lambda_{\min}(\mathcal{H}(t)) \geq \min\{\underline{\delta}, \lambda_{\min}(\mathcal{H}(T_e))\} > 0$ for $\forall t \geq T_e$.

Proof: Define the following variable related to the real symmetric $\mathcal{H}(t)$ provided in (42):

$$\Lambda^T\mathcal{H}(t)\Lambda = \int_0^t \left[\sqrt{K(t,\tau)}\Lambda^T\sigma_1(\tau)\right]^2 d\tau \quad (47)$$

where $\Lambda \in \mathbb{R}^N$ is the corresponding codimensional column vector and $0 < K(t, \tau) \leq 1$. Obviously, $\mathcal{H}(t) \geq 0$ for $\forall t \geq 0$ owing to $\Lambda^T\mathcal{H}(t)\Lambda \geq 0$ for any Λ and $t \geq 0$. If σ_1 fulfills FE, then $\int_{T_s}^{T_e} \sigma_1(\tau)\sigma_1^T(\tau)\,d\tau > 0$. Resorting to Lemma 1, $\int_0^t \sigma_1(\tau)\sigma_1^T(\tau)\,d\tau > 0$ for $\forall t \geq T_e$, which is equivalent to the nonexistence of $\Lambda \neq 0$ such that $\Lambda^T\sigma_1(\tau) \equiv 0$ for $\forall\tau \in [0, t]$ with $t \geq T_e$. Since $\sqrt{K(t,\tau)} > 0$, one can conclude that $\mathcal{H}(t) > 0$ for $t \geq T_e$. Because $\ell(t)$ is selected by $\lambda_{\min}(\mathcal{H}(t))$, $\ell(t) = 0$ if $\delta \leq \underline{\delta}$ at all time, meaning $\lambda_{\min}(\mathcal{H}(t)) \geq \lambda_{\min}(\mathcal{H}(T_e)) > 0$ for $\forall t \geq T_e$. Then, considering $\ell(t) > 0$, i.e., $\delta > \underline{\delta}$, one can obtain that $\ell(t)$ will reduce the past data to maintain the boundedness of $\mathcal{H}(t)$ and further renders $\lambda_{\min}(\mathcal{H}(t)) \geq \underline{\delta}$. To sum up, $\lambda_{\min}(\mathcal{H}(t)) \geq \min\{\underline{\delta}, \lambda_{\min}(\mathcal{H}(T_e))\} > 0$ for $\forall t \geq T_e$. ■

Theorem 5: Consider that Assumption 2 can be satisfied. Let $0 < \varpi \leq 1$ and $\bar{E}_h = \dot{\mathcal{H}}^{-1}\widetilde{\varepsilon}_H + \mathcal{H}^{-1}\dot{\widetilde{\varepsilon}}_H$. Then, the adaptive law (46) can ensure that the weight error $\tilde{W}$ decays to a small zone near the origin within a fixed arriving time T_w.

Proof: Based on (45), one gets $\mathcal{G} = -\mathcal{H}\tilde{W} + \widetilde{\varepsilon}_H$ with $\|\widetilde{\varepsilon}_H\| \leq \bar{\varepsilon}_H$. Resorting to Theorem 4, the minimum eigenvalue δ of the $\mathcal{H}(t)$ meets $\delta > 0$. Moreover, from (40), the matrix $\mathcal{H}(t)$ and its derivative $\dot{\mathcal{H}}$ are limited owing to the design of forgetting factor $\mathcal{K}(t)$. Therefore, $\bar{E}_h$ is bounded. Multiplying both sides of (45) by the matrix $\mathcal{H}^{-1}$ yields $\mathcal{H}^{-1}\mathcal{G} = -\tilde{W} + \mathcal{H}^{-1}\widetilde{\varepsilon}_H$, and its differential is $(d/dt)(\mathcal{H}^{-1}\mathcal{G}) = \dot{\hat{W}} + \bar{E}_h$. Next, define a Lyapunov function as

$$V_w = \mathcal{G}^T\mathcal{H}^{-1}\Gamma_W^{-1}\mathcal{H}^{-1}\mathcal{G}/2. \quad (48)$$

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According to (46) and Lemma 2, the differential of V_w is

$$\begin{aligned}\dot{V}_w &= \mathcal{G}^T\mathcal{H}^{-1}\Gamma_W^{-1}\frac{\partial\mathcal{H}^{-1}\mathcal{G}}{\partial t} = \mathcal{G}^T\mathcal{H}^{-1}\Gamma_W^{-1}\left(\dot{\hat{W}}+\dot{\bar{E}}_h\right)\\ &= -\mathcal{G}^T\mathcal{H}^{-1}\left(\frac{k_{W1}\mathcal{H}^T\mathcal{G}}{\sqrt{\|\mathcal{G}\|^2+\sigma_W^2}}+k_{W2}\mathcal{H}^T\mathcal{G}\mathcal{G}^T\mathcal{G}-\Gamma_W^{-1}\bar{E}_h\right)\\ &\le -\frac{(k_{W1}-\lambda_{\max}(\Gamma_W^{-1})\|\mathcal{H}^{-1}\bar{E}_h\|)\sqrt{2}\lambda_{\min}(\mathcal{H})}{\sqrt{\lambda_{\max}(\Gamma_W^{-1})}}V_w^{0.5}\\ &\quad -\frac{4k_{W2}(\lambda_{\min}(\mathcal{H}))^4}{(\lambda_{\max}(\Gamma_W^{-1}))^2}V_w^2+\sigma_{V_w}\\ &\le -\frac{(k_{W1}\delta-\lambda_{\max}(\Gamma_W^{-1})\|\bar{E}_h\|)\sqrt{2}}{\sqrt{\lambda_{\max}(\Gamma_W^{-1})}}V_w^{0.5}-\frac{4k_{W2}\delta^4}{(\lambda_{\max}(\Gamma_W^{-1}))^2}\\ &\quad\times(1-\varpi)V_w^2-\varpi\frac{4k_{W2}\delta^4}{(\lambda_{\max}(\Gamma_W^{-1}))^2}V_w^2+\sigma_{V_w}\end{aligned}\tag{49}$$

where $\sigma_{V_w} = k_{W1}\sigma_W$ is a positive number. If $-4\varpi k_{W2}\delta^4/(\lambda_{\max}(\Gamma_W^{-1}))^2V_w^2+\sigma_{V_w}\le 0$ holds, one obtains $\dot{V}_w \le -[(k_{W1}\delta-\lambda_{\max}(\Gamma_W^{-1})\|\bar{E}_h\|)\sqrt{2}/\sqrt{\lambda_{\max}(\Gamma_W^{-1})}]V_w^{0.5} - [4k_{W2}\delta^4/(\lambda_{\max}(\Gamma_W^{-1}))^2](1-\varpi)V_w^2$. Utilizing Lemma 3, the term $\mathcal{G}(t)$ will converge to origin in a fixed settling time T_w, with T_w being

$$T_w \le \frac{\sqrt{2\lambda_{\max}(\Gamma_W^{-1})}}{k_{W1}\delta-\lambda_{\max}(\Gamma_W^{-1})\|\bar{E}_h\|}+\frac{(\lambda_{\max}(\Gamma_W^{-1}))^2}{4k_{W2}(1-\varpi)\delta^4}.\tag{50}$$

Then, revisiting $\mathcal{G} = -\mathcal{H}\tilde{W}+\bar{\varepsilon}_H$, one has $\lim_{t\to T_w}\|\tilde{W}\| = \|\mathcal{H}^{-1}\bar{\varepsilon}_H\| \le \bar{\varepsilon}_H/\lambda_{\min}(\mathcal{H}) = \bar{\varepsilon}_H/\delta$. Thus, the settling time of weight errors is associated with the learning gain k_{W1}, k_{W2}, Γ_W, excitation level δ, and residual error $\bar{\varepsilon}_H$. The ultimate steady-state region is determined by δ and $\bar{\varepsilon}_H$. ■

Corollary 1: The control policies u_p and u_e provided by (37) approximate the optimal action pair (u_p^*, u_e^*) within the small bounds $\mathfrak{T}_1$ and $\mathfrak{T}_2$, respectively.

Proof: Restating $u_p^* = -\lambda_p\tanh(\Phi_p^*)$ with $\Phi_p^* = (1/2)R_p^{-1}G^T(\zeta)\nabla V^*(\zeta)$ and applying mean-value theorem, one has

$$u_p^* = -\lambda_p\tanh(\Phi_p)+\epsilon_{u_p^*}\tag{51}$$

where $\Phi_p = (1/2)R_p^{-1}G^T(\zeta)\nabla^T\Psi(\zeta)W$ and $\epsilon_{u_p^*} = -(\lambda_p/2R_p)(1-\tanh^2(a))G^T(\zeta)\nabla\epsilon$ with $a\in\mathbb{R}$ chosen between Φ_p and Φ_p^*. Let $\Phi_{\hat{p}} = (1/2)R_p^{-1}G^T(\zeta)\nabla^T\Psi(\zeta)\hat{W}$, $\mathcal{A}(\Phi_p) = \tanh(\Phi_p(\zeta))$, and $\mathcal{A}(\Phi_{\hat{p}}) = \tanh(\Phi_{\hat{p}}(\zeta))$. Based on the Taylor series, one gets

$$\begin{aligned}\mathcal{A}(\Phi_p)-\mathcal{A}(\Phi_{\hat{p}}) &= \frac{\partial\mathcal{A}(\Phi_{\hat{p}})}{\partial\Phi_{\hat{p}}}\big(\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta)\big)\\ &\quad+\mathcal{O}\Big((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2\Big)\\ &= \frac{1}{2}R_p^{-1}\Big[1-\tanh^2(\Phi_{\hat{p}}(\zeta))\Big]G^T(\zeta)\nabla^T\Psi\tilde{W}\\ &\quad+\mathcal{O}\Big((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2\Big)\end{aligned}\tag{52}$$

where $\mathcal{O}((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2)$ denotes the remainder higher-order terms. Next, by utilizing (52), we can derive $\mathcal{O}((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2) = \mathcal{A}(\Phi_p)-\mathcal{A}(\Phi_{\hat{p}})+(1/2)R_p^{-1}[\tanh^2(\Phi_{\hat{p}}(\zeta))-1]G^T(\zeta)\nabla^T\Psi\tilde{W}$. Due to $\|\tanh(\cdot)\|\le 1$, one obtains the fact of

$$\left\|\mathcal{O}\Big((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2\Big)\right\| \le c_1+c_2\|\tilde{W}\|$$

where c_1 and c_2 represent computable positive numbers. Then, by (37), (51), and (52), we have

$$\begin{aligned}\left\|u_p-u_p^*\right\| &= \left\|\lambda_p\big[\tanh(\Phi_{\hat{p}})-\tanh(\Phi_p)\big]+\epsilon_{u_p^*}\right\|\\ &= \Big\|\lambda_pR_p^{-1}\Big[1-\tanh^2(\Phi_{\hat{p}}(\zeta))\Big]G^T(\zeta)\nabla^T\Psi\tilde{W}/2\\ &\quad+\lambda_p\mathcal{O}\Big((\Phi_p(\zeta)-\Phi_{\hat{p}}(\zeta))^2\Big)+\epsilon_{u_p^*}\Big\| \le \lambda_pc_1\\ &\quad+\Big(\lambda_pR_p^{-1}G_M\bar{\Psi}/2+\lambda_pc_2\Big)\|\tilde{W}\|+\lambda_pG_M\bar{\epsilon}/2R_p.\end{aligned}\tag{53}$$

Using Theorem 5, $\tilde{W}$ is bounded within a fixed time T_w with ultimate bound $\bar{\varepsilon}_H/\delta$. Therefore, it yields

$$\begin{aligned}\left\|u_p-u_p^*\right\| &\le \bar{\varepsilon}_H\Big(\lambda_pR_p^{-1}G_M\bar{\Psi}/2+\lambda_pc_2\Big)/\delta\\ &\quad+\lambda_pc_1+\lambda_pG_M\bar{\epsilon}/2R_p \triangleq \mathfrak{T}_1.\end{aligned}\tag{54}$$

Similarly, we can find

$$\begin{aligned}\left\|u_e-u_e^*\right\| &\le \bar{\varepsilon}_H\Big(\lambda_eR_e^{-1}K_M\bar{\Psi}/2+\lambda_ec_4\Big)/\delta\\ &\quad+\lambda_ec_3+\lambda_eK_M\bar{\epsilon}/2R_e \triangleq \mathfrak{T}_2\end{aligned}\tag{55}$$

where c_3 and c_4 denote positive numbers. ■

Theorem 6: Consider a nonlinear system (6) for pursuit–evasion game with associated HJI equation (20). Let Assumptions 1 and 2 hold and let the near-optimal policy be provided by (37). Meanwhile, the NN learning law is generated by (46). Then, the system is ensured to be uniformly ultimately bounded.

Proof: Define a Lyapunov function $V^*(\zeta)$ given in (12). Taking the differential of $V^*(\zeta)$ along the trajectory $\dot{\zeta} = F(\zeta)+G(\zeta)u_p+K(\zeta)u_e$ and observing (20), one has

$$\begin{aligned}\dot{V}^*(\zeta) &= \nabla^TV^*(F(\zeta)+G(\zeta)u_p+K(\zeta)u_e)\\ &= \nabla^TV^*(F+Gu_p^*+Ku_e^*)+\nabla^TV^*G(u_p-u_p^*)\\ &\quad+\nabla^TV^*K(u_e-u_e^*)\\ &= -\zeta^TQ\zeta-2\int_0^{u_p^*}(\tanh^{-1}(\nu/\lambda_p))R_pd\nu\\ &\quad+2\int_0^{u_e^*}(\tanh^{-1}(\varpi/\lambda_e))R_ed\varpi+\nabla^TV^*G(u_p\\ &\quad-u_p^*)+\nabla^TV^*K(u_e-u_e^*).\end{aligned}\tag{56}$$

Based on the boundedness of ∇^TV^* from (33), and utilizing Theorem 5 and Corollary 1, it renders

$$\begin{aligned}\left\|\nabla^TV^*G(u_p-u_p^*)\right\| &\le G_M(\bar{\Psi}\bar{W}+\bar{\epsilon})\mathfrak{T}_1\\ \|\nabla^TV^*K(u_e-u_e^*)\| &\le K_M(\bar{\Psi}\bar{W}+\bar{\epsilon})\mathfrak{T}_2\end{aligned}\tag{57}$$

Resorting to Theorems 2 and 3, we can obtain that $-\zeta^TQ\zeta - 2\int_0^{u_p^*}(\tanh^{-1}(\nu/\lambda_p))R_pd\nu+2\int_0^{u_e^*}(\tanh^{-1}(\varpi/\lambda_e))R_ed\varpi(\zeta) \le -\zeta^TQ\zeta - L_c(\zeta)$ with $L_c(\zeta)\ge 0$. Thus, one derives

$$\dot{V}^*(\zeta) \le -\lambda_{\min}(Q)\|\zeta\|^2+\mathfrak{T}_V\tag{58}$$

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TABLE I
PARAMETER DESCRIPTIONS

Parameter		Value
Model parameters	Pursuer	$\bar{v}_p = 7(m/s), \bar{\omega}_p = 0.7(rad/s)$
	Evader	$\bar{v}_e = 5(m/s), \bar{\omega}_e = 0.6(rad/s)$
Fixed-time convergent RL with FE	Cost function	$Q = 5 \times I_2, \lambda_p = 0.7,$ $\lambda_e = 0.6, R_p = 1.0, R_e = 1.5$
	Critic NN	$\Psi(\zeta) = [\zeta_1^2, \zeta_1\zeta_2, \zeta_2^2]^T, N = 3,$ $\Gamma_W = 10 \times I_3, k_{W1} = 10,$ $k_{W2} = 200, k_0 = 1, \bar{\delta} = 0.1, \underline{\delta} = 0.001$

where $\mathfrak{T}_V \triangleq G_M(\bar{\Psi}\bar{W} + \bar{\epsilon})\mathfrak{T}_1 + K_M(\bar{\Psi}\bar{W} + \bar{\epsilon})\mathfrak{T}_2$. Therefore, it follows from (58) that $\dot{V}^*(\zeta) < 0$ as long as $\zeta(t)$ locates outside the compact set

$$\Omega_\zeta = \left\{\zeta : \|\zeta\| \le \sqrt{\mathfrak{T}_V/\lambda_{\min}(Q)}\right\}. \tag{59}$$

Thus, the relative motion dynamics system remains stable and exhibits ultimate boundedness during the learning process. ■

Remark 5: In the prevailing design [19], [20], [21], [22], [23], [24] of the learning law, it is desired to set $\hat{W}$ to minimize the current Hamiltonian estimation error $E(\hat{W}(t)) \triangleq (1/2)\delta_e(t)^2$, $t \ge 0$. Then, the normalized gradient descent is used to update the tuning law of the critic NN asymptotically, i.e., $\dot{\hat{W}} = -\Gamma_1[\sigma_1/(\sigma_1^T\sigma_1 + 1)^2]\delta_e = \Gamma_1[1/(\sigma_1^T\sigma_1 + 1)^2](\sigma_1\sigma_1^T\tilde{W} + \sigma_1\varepsilon_H)$, with $\Gamma_1 > 0$. However, the rank of regression variable $\sigma_1\sigma_1^T$ is always no greater than 1 since all rows are linearly dependent. Thus, strict PE is needed to ensure that the input data exhibit sufficient variability and information to guarantee error convergence, which is extremely challenging. To remove this restriction, some elegant methods [29], [30] are established by storing the discrete data stacks from the past measurements. Unfortunately, this design requires the adoption of complex algorithms for data selection and determining the linear independence condition. Moreover, the asymptotic tuning law results in sluggish learning efficiency. To overcome the above limitations, the incoming data are first collected to form the auxiliary information matrix $\mathcal{H}$ and variable $\mathcal{M}$, which is then evaluated against a selected metric to determine and retain the highest quality matrix. As a result, the auxiliary matrix can be structured to attain full rank after finite but not PE, promoting rapid convergence. Moreover, a fixed-time learning law is presented to significantly accelerate the convergence of weights to desired values.

V. SIMULATIONS

In this section, we establish a set of simulations to verify the effectiveness and advantage of the proposed methodology. All simulations are conducted using the MATLAB/Simulink platform with a sampling interval of 0.01 s. The involved parameters are listed in Table I.

A. Effectiveness Verification

Consider a pursuit–evasion game involving two vehicles. Both the pursuer and the evader are assumed to maneuver at their respective maximum speeds, with the pursuer aiming

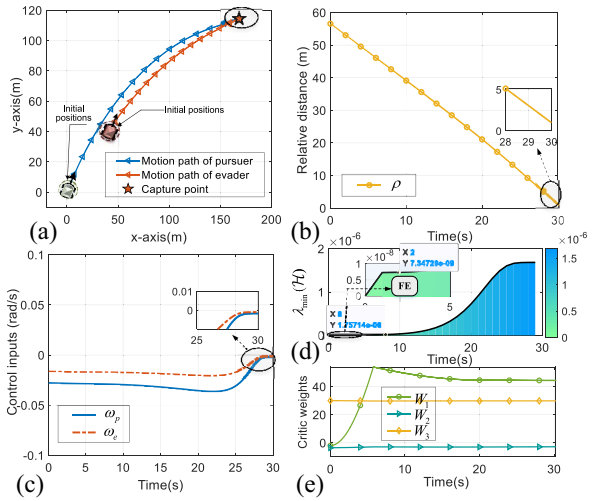


Fig. 3. Evolutions of the pursuit–evasion game for two players under case 1. (a) Interaction trajectories. (b) Episode of the relative distance between the pursuer and evader. (c) Control quantities u_p and u_e. (d) Excitation strength of the auxiliary matrix based on FE. (e) Estimated weights $\hat{W}$ of critic NN.

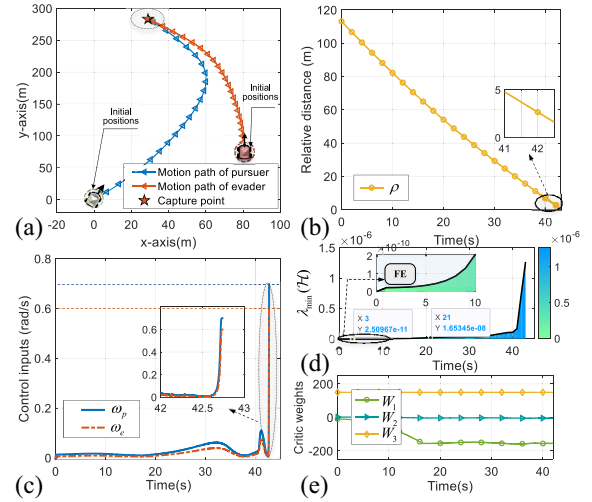


Fig. 4. Evolutions of the pursuit–evasion game for two players under case 2. (a) Interaction trajectories. (b) Episode of the relative distance between the pursuer and evader. (c) Control quantities u_p and u_e. (d) Excitation strength of the auxiliary matrix based on FE. (e) Estimated weights $\hat{W}$ of critic NN.

to respond optimally to the evader's maneuvers to achieve capture. Consequently, the pursuer and the evader adopt the optimal strategy pair (37). To satisfy the sufficient conditions for capture as outlined in Theorems 2 and 3, the parameters for the pursuer are specified in Table I. We account for realistic physical constraints by defining the capture radius as 1 m. Then, we examine the evolution of the pursuit–evasion game under three distinct initial conditions, designated as Cases 1–3.

Figs. 3–5 illustrate the game procedure under the optimal strategies. As shown in Figs. 3(a)–5(a), the pursuer can capture the target within a finite time under any initial conditions, provided the target is capturable. The corresponding relative distance is depicted in Figs. 3(b)–5(b), which shows that the pursuer can progressively reduce the range until the mission is accomplished. The smooth and bounded control actions

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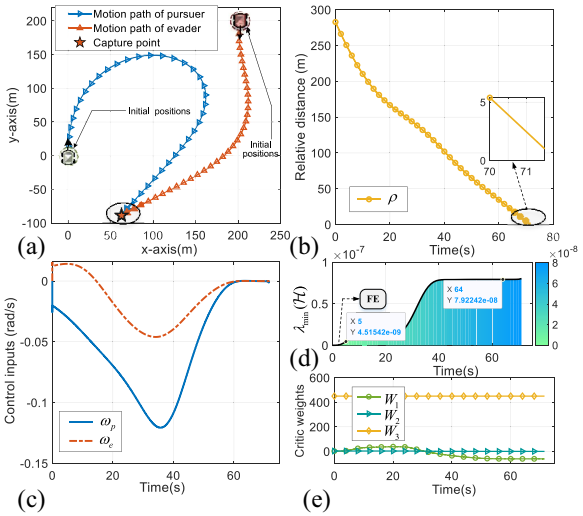


Fig. 5. Evolutions of the pursuit–evasion game for two players under case 3. (a) Interaction trajectories. (b) Episode of the relative distance between the pursuer and evader. (c) Control quantities u_p and u_e. (d) Excitation strength of the auxiliary matrix based on FE. (e) Estimated weights $\hat{W}$ of critic NN.

TABLE II
THREE GROUPS OF DIFFERENT CONDITIONS

Case	Initial positions and attitudes $[x_i, y_i, \psi_i]$		Results	Capture time
1	Pursuer	$[0, 0, \pi/3]$	Capture	30.1s
	Evader	$[40, 40, \pi/3]$		
2	Pursuer	$[0, 0, \pi/4]$	Capture	42.4s
	Evader	$[80, 80, \pi/2]$		
3	Pursuer	$[0, 0, \pi/2]$	Capture	71.3s
	Evader	$[200, 200, 3\pi/2]$		

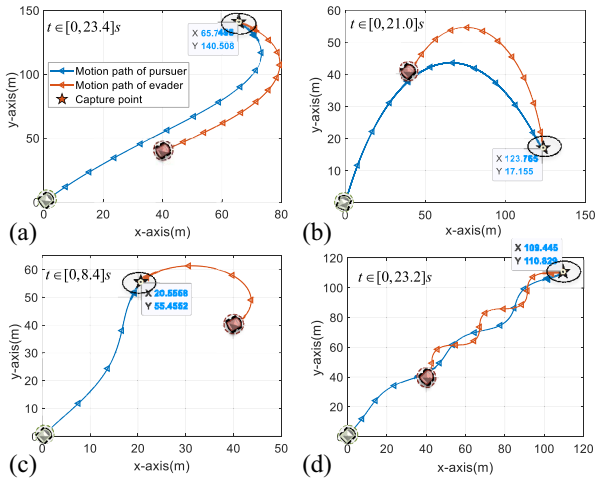


Fig. 6. Interaction trajectories based on the pursuer's optimal policy $u_p^* \triangleq$ (16) and the evader's nonoptimal baseline policies under case 1: (a) $u_p^* \triangleq$ (16), $u_e \triangleq 0.6\sin(t/100)$, (b) $u_p^* \triangleq$ (16), $u_e \triangleq -0.1$, (c) $u_p^* \triangleq$ (16), $u_e \triangleq 0.4$, and (d) $u_p^* \triangleq$ (16), $u_e \triangleq 0.2\sin(3t/2)$.

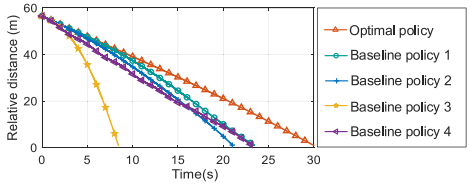


Fig. 7. Evolutions of the relative distances between the pursuer and evader with different policies under case 1.

are illustrated in Figs. 3(c)–5(c). Figs. 3(d)–5(d) provide the conditions for the convergence of NN weights, where the eigenvalues of the matrix are greater than zero after a short time, i.e., $\lambda_{\min}(\mathcal{H}(t)) > 0$, indicating that the FE condition can be well satisfied. Finally, Figs. 3(e)–5(e) illustrate the estimated weights of the critic NN, and one can find that after undergoing excitation, the weights can be stabilized, indicating that both agents reach an agreement on the game behavior. Table II summarizes the game outcomes. It is evident that when the pursuer satisfies sufficient conditions, the alteration of the initial states does not alter the eventual capture outcome, but it can influence the performance indexes of both players, leading to different game behaviors, which is consistent with the verified theoretic outcomes shown in Theorems 1–3. Thus, the feasibility of the proposed strategy is validated.

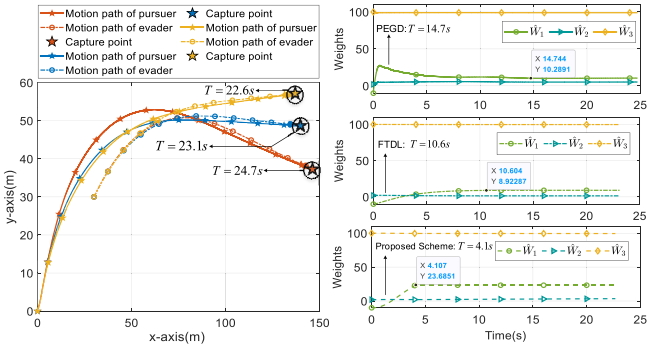
B. Comparative Analysis

To demonstrate the effectiveness of the proposed game-theoretic approach and characterize the interaction behaviors, we compare the results when the evader adopts either the optimal strategy (17) or several predefined nonoptimal baseline strategies. Specifically, the considered baseline policies include sinusoidal maneuvers with varying frequencies and maneuvers with constant angular rates, defined as follows: Baseline 1: $u_e = 0.6\sin(t/100)$; Baseline 2: $u_e = -0.1$; Baseline 3: $u_e = 0.4$; and Baseline 4: $u_e = 0.2\sin(3t/2)$. Then, the coupled HJI equation (20) reduces to the Hamilton–Jacobi–Bellman (HJB) equation by setting $L_e(u_e) = 0$, representing the pursuer's unilateral optimal control problem in this case. The pursuer's optimal policy is derived according to (16). All related parameters and simulation conditions remain consistent with those in case 1. The simulation results are shown in Figs. 6 and 7. We can observe that when the evader adopts its game-theoretic optimal strategy, it takes longer for the pursuer to intercept the evader compared to when the evader uses an irrational baseline strategy. This validates the efficacy of the proposed method in predicting the evader's optimal maneuvers.

Furthermore, to emphasize the benefits of the proposed novel residual design structure, we conducted comparative simulations against the two prevailing critic-driven RL methods: 1) the PE-based gradient descent method [19] (abbreviated as PEGD) and 2) the finite-time data-driven learning method [31] (abbreviated as FTDL). To ensure fairness, all conditions and parameters are kept consistent, except for replacing the learning law (46). For a quantitative evaluation, we formulate a cost function: $V_c(t) = \int_0^t (v_p^2 + \rho^2) d\tau$. Additionally, to ensure the PE condition of the comparative scheme, a small probing noise $n(t) = 0.05\exp(-0.05t)(\sin(0.01t) + \sin(0.1t) + \sin(t))$ is added to the control input for $t \in [0, 3]$ s. The comparative outcomes

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(a) (b)

Fig. 8. Evolutions of the pursuit–evasion game. (a) Interaction trajectories: the solid line denotes the pursuer's path, and the dotted line denotes the evader's path. The red line corresponds to PEGD [19], the blue line to FTDL [31], and the yellow line to the proposed method. (b) Estimated weight $\hat{W}$.

TABLE III
COMPARISON OF DIFFERENT RL METHODS

Index \ Scheme	PEGD [19]	FTDL [31]	Proposed scheme
Cost V_c	19529.6	17832.5	16636.7
Capture time	24.7s	23.1s	22.6s
Weight settling time	14.7s	10.6s	4.1s

are shown in Fig. 8, revealing that all strategies successfully achieve target capture. Differently, the proposed scheme achieves a shorter capture time compared with the PEGD and FTDL. This improvement benefits from the relaxation of the stringent PE condition to FE condition and the adoption of a fixed-time learning rule, enabling faster acquisition of the optimal strategy pairs. As shown in Table III, the comparison of quantitative metrics demonstrates that the proposed method accelerates the convergence of NN weights by 6.5 s compared to the state-of-the-art FTDL method, under identical conditions, thereby reducing the capture time and lowering the capture cost.

VI. CONCLUSION

This article studied the optimal solution to the pursuit–evasion intercept issue, considering the input saturation imposed on both players along with kinematic constraints. The existence of Nash equilibrium under these input constraints was proven, and it was demonstrated that the game process would terminate within a finite time. To solve the HJI equation, a critic-only NN was introduced. By constructing a novel residual design scheme based on current and historical data and employing a fixed-time learning mechanism, a new learning algorithm was proposed, which not only eliminated the PE condition but also achieved fixed-time convergence. Simulations revealed the interaction process between the pursuer and the evader, validating the feasibility and advantages of the proposed method.

The proposed method has the following limitations. First, it does not account for unreachable regions, such as obstacles or boundaries, which could affect the pursuer's ability to intercept the evader. Future work should integrate obstacle avoidance techniques for improved practicability. Second, the assumption of perfect rationality may not reflect real-world behaviors, where players may exhibit bounded rationality. Future research will incorporate the level-k thinking framework to model these behaviors and explore their impact. Additionally, multiple-player scenarios and real-world experiments will be considered to further validate the approach.

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